Use-Case Flow Specification

UC-03 — Transfer Solar Credits

Primary Actor: Prosumer Resident
Supporting Actor: Utility Billing System
Goal: Transfer surplus solar credits to a registered neighbor's account.

Preconditions

1. The Prosumer Resident is registered with the Community Solar Credit Allocation Manager.
2. The Prosumer Resident is authenticated in the system.
3. The recipient is a registered participating neighbor.
4. The Prosumer Resident has sufficient available solar credits for the requested transfer.

Postconditions

Success:

• Credits are deducted from the sender.
• Credits are added to the recipient.
• The transfer is recorded.
• Applicable recipient billing information is updated.

Failure:

• No credits are deducted or added.
• The failed attempt is reported to the Prosumer Resident.

Main Success Scenario

1. The Prosumer Resident selects Transfer Solar Credits.
2. The system displays the prosumer's available solar credit balance.
3. The Prosumer Resident enters the registered neighbor's account and the number of credits to transfer.
4. The system validates the recipient account and requested transfer amount.
5. The system verifies that the Prosumer Resident has sufficient available credits.
6. The system confirms the transfer details with the Prosumer Resident.
7. The Prosumer Resident confirms the transfer.
8. The system deducts the specified credits from the Prosumer Resident's account.
9. The system adds the transferred credits to the recipient's account.
10. The system records the transaction.
11. The system updates the applicable billing information.
12. The system displays a transfer-success confirmation to the Prosumer Resident.
13. The use case ends successfully.

Alternate Flow — Insufficient Credit Balance

1. The requested transfer exceeds the available credit balance.
2. The system rejects the transfer and displays an insufficient-balance message.
3. The Prosumer Resident may enter a smaller amount and retry.
4. If they do not retry, neither account balance is modified.